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Small satellites for global coverage: Potential and limits

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ABSTRACT

There is an increasing need for Earth Observation (EO) missions to meet the information requirements in connection with Global Change Studies. Small and cost-effective missions are powerful tools to flexibly react to information requirements with space-borne solutions. Small satellite missions can be conducted relatively quickly and inexpensively by using commercial off-the-shelf-technologies, or they can be enhanced by using advanced technologies. A new class of advanced small satellites, including autonomously operating "intelligent" satellites may be created, opening new fields of application. The increasing number of small satellites and their applications drive developments in the fields of small launchers, small ground station networks, cost-effective data distribution methods, and cost-effective management and quality assurance procedures. There are many advantages of small satellite missions, like more frequent mission opportunities, a faster return of data, larger variety of missions, more rapid expansion of the technical and/or scientific knowledge base, greater involvement of small industry, feasibility by universities and others.

This paper deals with general trends in the field of small satellite missions for Earth observation. Special attention is given to the potential of spatial, spectral, and temporal resolution of small satellite based systems. Examples show small satellites offer also the unique possibility to install affordable constellations to provide good daily coverage of the globe and/or allow us to observe dynamic phenomena.

The facts and examples given in this paper lead to the conclusion: Small satellites are already powerful tools for monitoring global, regional and local phenomena. In the future, their application spectrum will even broaden based on the ongoing development in many areas of technology and observation techniques.

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1. Introduction

Space-borne Earth observation has become a valuable tool in sustainable management and global change studies of the Earth because of its unique capability for acquiring measurements of various environmental conditions over large areas. The great need for Earth observation missions in order to improve the related data base contents is perhaps most clearly seen in the many current moves for international co-operation in the field of the environment where measurements from Earth observing satellites are an essential element. This is especially so where we need to acquire, analyze and use data documenting the condition of the Earth's resources and environment on a long-term (permanent) basis.

For instance, in 2008 the Group on Earth Observations, which currently numbers some 74 countries, the European Commission and 51 participating organizations, has concrete plans for its Global Earth Observation System of Systems. In 2008 the European Union's Space Council continued to advance Europe's space policy, reaffirming the need for rapid implementation of the Global Monitoring for Environment and Security (GMES) program.

From the space-borne remote sensing point of view, the only way to implement flexible space systems for security and prosperity of the people is to pursue activities to develop and operate cost-effective Earth observation missions in order to monitor the relevant geophysical phenomena globally.

The following sections deal with general facts and trends in the field of small satellite missions for Earth observation. Special attention is given to the potential of spatial, spectral, and temporal resolution of small satellite based systems. The capabilities of small satellites in terms of spatial and spectral resolution are close to what larger satellites can provide. Small satellites offer also the unique possibility to install affordable constellations to provide good daily coverage of the globe and/or allow us to observe various dynamic phenomena through their potential of increasing the

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temporal resolution. The examples show that small satellite based systems are well suited for monitoring global, regional and local phenomena.

The main obstacles small satellites face are the restrictions in mass and volume. That leads to restrictions in power consumption and data rates because of the physical size of components like solar panels and antennas. This applies also to application-specific instrument components like high resolution optics and radar antennas. But also in those fields of technology and application the development results are encouraging. Another important obstacle caused by restricted mass and volume is the focus on selected physical phenomena from a small satellite platform. But this may even turn out to become an advantage if we think of building complex Earth observation systems using a complex suite of instruments accommodated on a distributed small satellite platform system. In the case of the malfunction of one system, it can be easily and cost-effectively replaced. So, in the future the number of applications which will be covered by small satellite missions using active and passive remote sensing systems will be steadily increasing. This is a necessary process because the importance of remote sensing is increasing since we have to face an increasing world population and decreasing resources at the same time asking for an ever increasing number of cost-effective space systems to provide quality and timely information.

2. Small satellite missions: facts and trends

2.1. General facts

Small and cost-effective missions are powerful tools to flexibly react to information requirements with space-borne solutions. Small satellite missions can be conducted relatively quickly and inexpensively and provide increased opportunity for access to space. The spacecraft bus and instruments can be based either on optimized off-the-shelf systems, with little or no requirements for new technology, or on new high-technology systems. Thus a new class of advanced small satellites, including autonomously operating "intelligent" satellites may be created, opening new fields of applications for scientific purposes as well as operational, public and commercial services. Further milestones in the small satellite Earth observation mission development are the availability and improvement of small launchers, the development of small ground station networks connected with rapid and cost-effective data distribution methods, and cost-effective management and quality assurance procedures.

For about two decades, small satellites using off-the-shelf technologies or missions focused on specific physical phenomena have also been perceived to offer an opportunity for countries with a modest research budget and little or no experience in space technology, to enter the field of space-borne Earth observation and its applications. Small satellite technology is a major means to bring within the reach of every country the opportunity to operate small satellite Earth observation missions and utilize the data effectively at low cost, as well as to develop and build application-driven missions. It provides the opportunity to conduct or participate in Earth observation missions using small, economical satellites, and associated launchers, ground stations, data distributions structures, and space system management approaches.

One of the possible approaches of ending up with a small satellite system is taking full advantage of the ongoing technology developments leading to further miniaturization of engineering components, development of micro-technologies for sensors and instruments which allow designing dedicated, well-focused Earth observation missions. At the extreme end of miniaturization, the integration of micro-electromechanical systems (MEMS) with microelectronics for data processing, signal conditioning, power

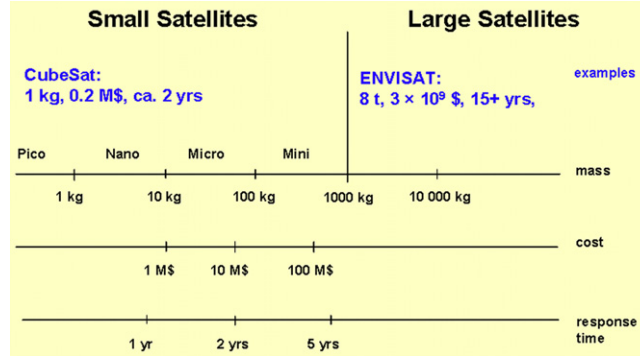

Fig. 1. Some features of small satellites.

conditioning, and communications leads to the concept of application-specific integrated micro-instruments (ASIM). These micro- and nano-technologies have led to the concepts of nano- and pico-satellites, constructed by stacking wafer-scale ASIMs together with solar cells and antennas on the exterior surface, enabling the concept of space sensor webs.

The situation in the field of small satellite missions for Earth observation has matured in the last ten years. This may be, for instance, observed from the topics and the quality of contributions to the series of conferences taking place in Berlin, Logan, at the annual International Astronautical Congress, or conducted by space agencies like ESA or CNES. Also the Small Satellite Workshops of ISPRS's Commission I give evidence for this maturity.

But what exactly is a small satellite? The International Academy of Astronautics (IAA) proposed a simplified definition (Sandau, 2006). This definition is reflected in Fig. 1 in conjunction with additional features which are essential when discussing small satellite characteristics like cost and response time. The performance issue is covered in the subsequent subsections.

At UNISPACE III (Background Paper, 1998), the costs of developing and manufacturing a typical mini-satellite were indicated to be US$ 5–20 million, while the cost of a micro-satellite was correspondingly US$ 2–5 million. The cost of a nano-satellite could be below US$ 1 million (prices as of 1999). Whereas the development and production time for large satellites is observed to be 15+ years, the corresponding time for minis should be 3–5 years, for micros 1.5 years, for nanos about 1 year, and for picos less than 1 year. Of course, cost and response time figures are to be considered ballpark figures. They are based on the usage of state-of-the-art technology by professional teams. They may deviate considerably if key technology is to be developed and/or the implementation teams are at the beginning of the learning curve. Fig. 1 is complemented by two examples showing the edges of the feature ranges. From the customer's point of view, the most important feature of small satellites is its performance. This aspect is covered in the following subsections.

From Fig. 1 we can learn that, roughly, the smaller the satellite the less the cost and the response time. This is a strong motivation to try to go for small satellite missions. The IAA study (Sandau, 2006) presented the state of the art of small satellite missions and examined more factors that enable one to produce a cost-effective small satellite mission for Earth observation. One of the key enablers of designing a cost-effective mission is having the key expertise available. As the number of successfully space-faring nations grows, the pool of expertise available to meet the challenges of small missions grows.

The advantages of small satellite missions are:

– more frequent mission opportunities and therefore faster return of science and for application data

- larger variety of missions and therefore also greater diversification of potential users
- more rapid expansion of the technical and/or scientific knowledge base
- greater involvement of local and small industry.

Large satellite missions and small satellite missions are considered to be complementary rather than competitive. The large satellite missions are sometimes even a precondition for cost-effective approaches.

2.2. Trends

Small satellite missions are supported by several contemporary trends (Sandau, 2006):

- Advances in electronic miniaturization and associated performance capability;
- The recent appearance on the market of new small launchers (e.g. through the use of modified military missiles to launch small satellites);
- The possibility of "independence" in space (small satellites can provide an affordable way for many countries to achieve Earth observation and/or defense capability, without relying on inputs from the major space-faring nations);
- Ongoing reduction in mission complexity as well as in those costs associated with management; with meeting safety regulations etc.;
- The development of small ground station networks connected with rapid and cost-effective data distribution methods.

But of course, the trend to smaller satellites was and is supported by the improvements in diverse fields of technology such as for instance optics, mechanics and materials, electronics, signal processing, communications and navigation in addition to microelectronics. Mass, volume and power consumption of the spacecraft and the instruments followed the trend to miniaturization allowing at the same time significant increases of performance. These trends can be observed for passive optical space-borne systems as well as for active microwave systems, e.g. SAR (Synthetic Aperture Radar) systems. They all benefit from the technology improvements.

3. Resolution requirements for space-borne remote sensing

When we discuss the performance parameters, we address at first the resolution in terms of:

- Spatial resolution.
- Spectral resolution.
- Temporal resolution.

For space-borne sensors, further requirements may be considered concerning mass, volume and power consumption. These are essential features when it comes to the spacecraft design, especially when targeting small satellites. Figs. 2 and 3 show the very diverse requirements connected with the different remote sensing application fields (Roeser et al., 2000). Fig. 2 relates the spectral requirements to the GSD (Ground Sampling Distance) for different application fields. Fig. 3 depicts temporal resolution versus GSD for different application fields.

The range of spatial resolutions (GSD) covers centimeters to several hundred meters. The revisit time (temporal resolution) ranges from less than one hour to 10 years. The range of spectral resolution requirements starts with panchromatic only for topographic mapping and ends with hyperspectral resolution, for instance in the field of hydrology.

It is obvious: a small satellite system for remote sensing needs to focus on one of the application fields and, within that application field, on a specific task or group of tasks where feasible. With this background, when talking about a small satellite we are basically talking about one instrument or a complex instrument system optimized for a specific task and accommodated on a small satellite platform.

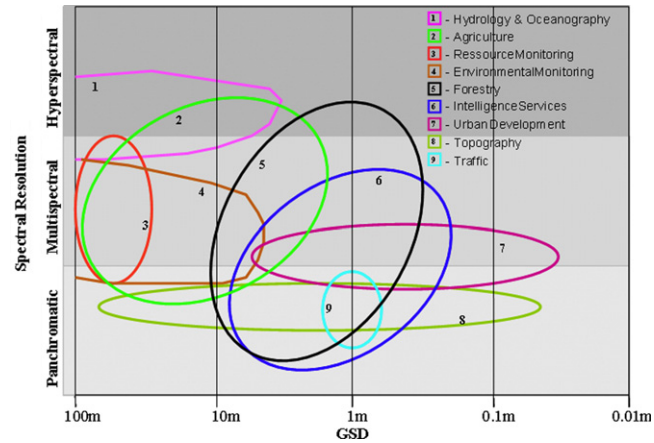

Fig. 2. Earth observation request: GSD versus spectral resolution.

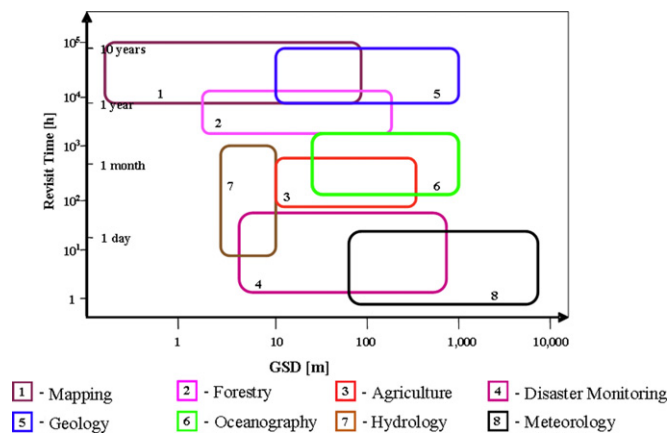

Fig. 3. Earth observation request: GSD versus revisit time.

4. Potential and limits

4.1. Capabilities of small satellites—optical payloads

Seeing the huge diversity in the three different resolution areas, spatial, spectral, and temporal (Fig. 4), we need to answer two questions (for optical remote sensing systems) for small satellites and their payloads:

- what are the limitations, and
- what are the strengths.

Spatial resolution

The spatial resolution is increasing, i.e., the GSD is decreasing. For example: the camera PIC-2 on the small satellite EROS-B from Israel provides a GSD of 0.70 m (EROS-B, 2006). EROS-B with a mass of 350 kg was launched on 25 April 2006 with a Russian START-1 launcher into a 500 km sun synchronous orbit (SSO).

Spectral resolution

Also the spectral resolution is increasing. As an example, we consider the hyperspectral imager CHRIS on the ESA funded PROBA satellite (PROBA, 2002). CHRIS, the 14 kg/9 W hyperspectral imager, has a GSD of 18 m and provides up to 19 out of a total of 62 spectral bands in the VIS/NIR spectral range (400–1000 nm).

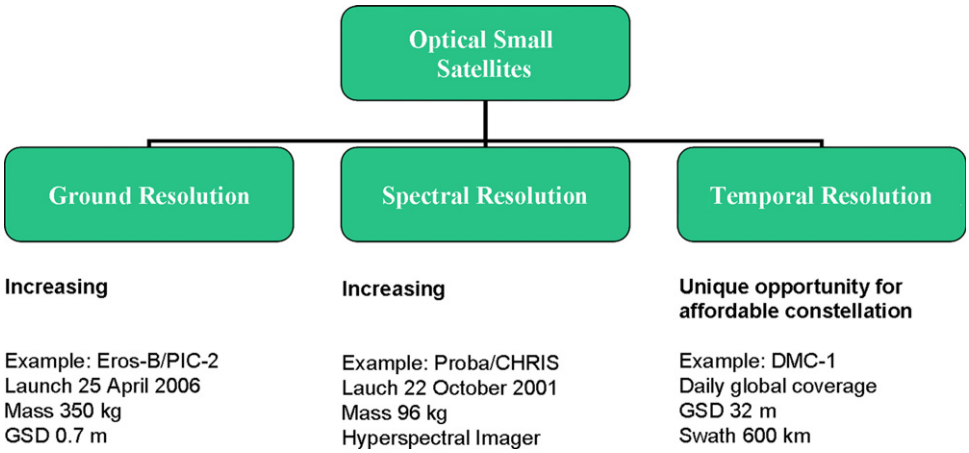

Fig. 4. Resolution aspects of optical payloads.

PROBA with a mass below 100 kg (so it is a micro-satellite) was launched into a 600 km sun synchronous orbit (SSO) on 22 October 2001 together with the DLR/Germany micro-satellite BIRD for forest fire detection and fire parameter assessment (Brieß et al., 2003), and the main payload TES (India) (TES, 2008) with the PSLV-C3 launcher from India. BIRD is described in more detail as one of the example systems in Section 5.1.1.

Temporal resolution

Small satellites provide a unique opportunity for affordable constellations. In this respect, small satellites can do things that are not practical with large satellites. At this point, the DMC (Disaster Monitoring Constellation) may serve as an example for a constellation of five small satellites (Da Silva Curiel, 2005). We will elaborate more on constellations and formations in Section 5.2.1 of this paper.

So the answer to the question posed at the beginning of this subsection is:

- for passive optical payloads there are limitations concerning the complexity or numbers of instruments on one spacecraft
- the strengths are the competitiveness with the performance of instruments on large spacecraft and the unique feature to build affordable constellations for improved daily coverage and temporal resolution.

4.2. Capabilities of small satellites—microwave payloads

Active microwave payloads have been typically associated with large and expensive platforms, mainly due to their mass and required power. However, radar altimeters, most useful for marine and ice sheet studies, are characterized by masses less than 100 kg and power requirements in the range 70–170 W. Thus, they are fully compatible with a small satellite approach. They have been flown on either large multi-payload platforms (typically in synergy with an SAR: e.g. SEASAT, ERS-1 and 2, ENVISAT) or on small platforms (e.g. Jason-1 and 2, Cryosat), often complemented with passive microwave radiometers.

Synthetic Aperture Radars, with huge mass and power requirements, have been flown by large satellites (Fig. 5). Nonetheless, if on the one hand the only examples of small satellites are given by TecSAR and SAR-Lupe, a five-satellite constellation (EO portal, 2009), it must be noted on the other hand that also "large" missions are approaching the 1 ton limit. SAR spatial resolutions can vary widely depending on the operating mode (ScanSAR, stripmap, spotlight, etc.). Fig. 6 shows the best achievable resolutions by spaceborne SARs, which have reached the few meter level, both in range and azimuth. Spatial resolutions are not related to satellite mass, as demonstrated by TecSAR and SAR Lupe (1 m × 1 m),

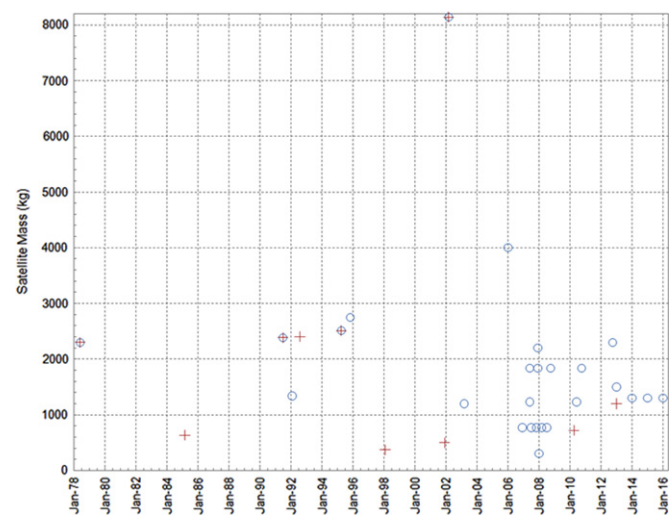

Fig. 5. Mass of satellites carrying radar altimeters (red crosses) or Synthetic Aperture Radars (blue circles).

whereas the selection of a small bus impacts the number of SAR operating modes. In fact, the necessity of reducing payload mass and power calls for simplified antenna solutions with respect to the most advanced active phase arrays (e.g. COSMO/SkyMed, TerraSAR-X, Radarsat-2, etc.), generally impacting ScanSAR (wide swath, reduced spatial resolution) capability. Reduced available average power also has an impact on payload duty cycle per orbit.

4.3. Limitations

Small satellite missions usually focus on one specific physical phenomenon to be investigated or monitored. In this context, the restrictions or limitations of small satellite missions in comparison to large complex missions are:

- In orbit lifetime restrictions because of the extended use of advanced technologies (in comparison to conventional satellite missions).
- Limited platform capacity for using instruments with high power consumption or high data rate requirements.
- Size limitations and platform stability limitations do not allow the use of large microwave antennas or long monolithic telescopes.
- Restricted options for instrument combinations on a single satellite platform because of the limited size and power capabilities of small satellites.

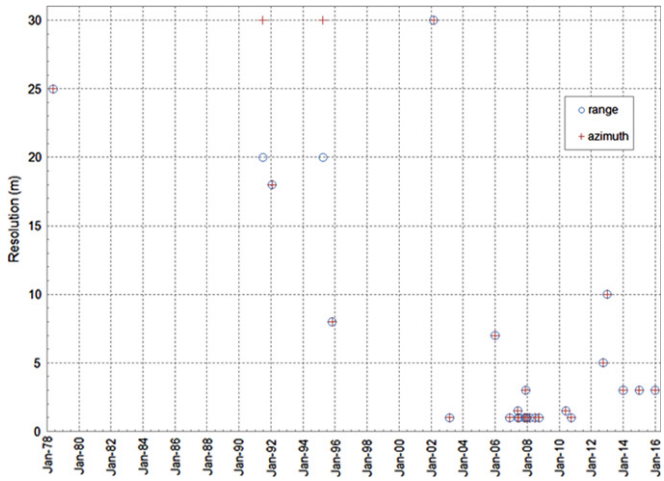

Fig. 6. Resolutions of spaceborne SARs.

For these reasons small satellite missions may be considered as supplements to conventional Earth observation missions.

5. Examples of the actual status and future developments

5.1. Systems for Earth observation and surveillance

For two decades on average two mini- or micro-satellites from many countries have been launched serving the needs of many user groups (Kramer, 2002). They represent also different approaches to make use of the specifics of small satellites. For instance, one interesting approach is used by SAC-C, Argentina's first satellite for Earth observation. It was injected into orbit on 2 November 2000, in NASA's morning train orbit and is used to make complex measurements from the small satellite platform, also using the data coming from Landsat-7, EO-1 and Terra. With this approach, SAC-C is part of a distributed system (see Section 5.2.2) using payloads on board different satellites synergistically rather than multiplying payloads for coverage enhancement (for constellations, see Section 5.2.1).

There are various interesting systems in orbit for which this subsection gives short descriptions of two examples exemplifying

- the efficiency of a micro-satellite optimized to deal with one physical phenomenon
- the efficiency of a micro-satellite with high agility to deal with local phenomena in real-time.

5.1.1. Hot spot detection and assessment

The micro-satellite BIRD (Bi-spectral InfraRed Detection) with its specialized payload is used to describe a possible approach for the detection and quantitative characterization of high-temperature events like vegetation fires on the Earth's surface. BIRD (see Fig. 7) was successfully piggyback launched by an Indian Polar Satellite Launch Vehicle (PSLV-C3) into a circular sun-synchronous orbit with an altitude of 572 km on 22 October 2001. A detailed description of the BIRD system is given in Brieß et al. (2003). Here we only show an example of BIRD performance (payload characteristics are given in Table 1). Fig. 8 gives a comparison of results derived from MODIS (currently the best satellite system to detect fires) and BIRD of the forest fire in Australia near Sydney on 5 January 2002. The difference is obvious. BIRD data even allow for providing such important parameters to fire fighting authorities like fire temperature, front length and front strength in kW/m (see Fig. 9 and Table 2).

Table 1
Characteristics of the BIRD main sensor payload (orbit altitude = 572 km).

	WAOSS-B	MWIR	TIR
Wavelength	600–670 nm 840–900 nm	3.4–4.2 μm	8.5–9.3 μm
Focal length	21.65 mm	46.39 mm	46.39 mm
Field of view	50°	19°	19°
f-number	2.8	2.0	2.0
Detector	CCD lines	CdHgTe arrays	CdHgTe arrays
Detector cooling	passive, 20 °C	Stirling, 80 K	Stirling, 80 K

Fig. 7. Micro-satellite BIRD, mass of spacecraft: 94 kg, mass of payload: 30.2 kg.

Fig. 10 illustrates the BIRD potential for fire detection during daytime and at night. The hotspot in the white circle in a daytime BIRD image in Fig. 10(a) was verified to correspond to a small fire with an area of 12 m^2 ($\sim 10^{-4}$ of the pixel area of the BIRD IR channels). Fig. 10(b) shows a small controlled wood fire with an area of only 4 m^2 (3×10^{-5} of the IR pixel area) arranged at night during a BIRD overflight (Zhukov et al., 2005).

5.1.2. Interactive surveillance payload systems

An interactive surveillance payload system is used in cases where the target is not clearly identified in advance, a search action is involved or a target has to be followed for a while. Such a payload system and the associated satellite platform have to be developed in very close collaboration.

The microsatellite DLR-TUBSAT may serve as an example for such an application (Schulz and Renner, 2000). It was a joint project of the Institute of Aerospace at the Technical University of Berlin (TUB) and the German Aerospace Centre (DLR). The microsatellite was launched on 26 May 1999 with the Indian Polar Spacecraft Launch Vehicle (PSLV) together with KITSAT-3 and the primary payload IRS-P4. It was developed as a test satellite in order to test the necessary technologies (camera, position and attitude measurement systems, attitude control, software for interaction with the ground station, etc). The intention of this development was also

- to provide a low-cost platform that can be configured and launched for a very specific mission within a short time (operational responsive space),
- to take advantage of their mobility of monitoring processes which change fast (within one day, for instance), for search actions or for following a target.

Micro-satellites have a high mobility due to their compactness and small mass which is essential for observing the following categories of events:

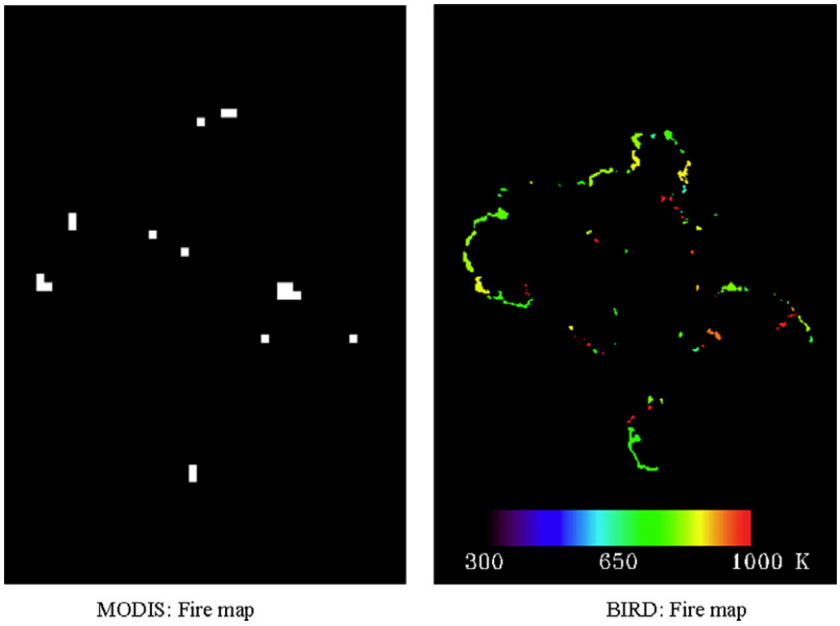

Fig. 8. Fire detection by MODIS and BIRD (Australia, 5 January 2002).

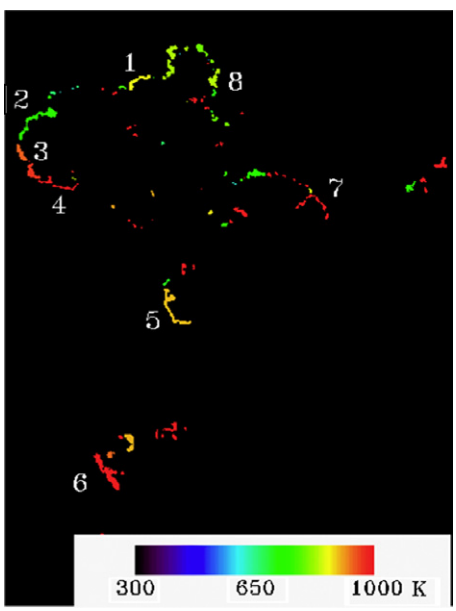

Fig. 9. Typical characteristics of fire fronts (BIRD, Australia, 5 January 2002).

- weather phenomena like hurricanes
- lightning or polar light
- spectacular fires
- volcano eruptions
- floods
- earthquakes
- ship, plane or railway accidents

or any other event for which for instance an authority would like to be informed with only a short delay. The area of interest is typically only a few square kilometers large and the spatial resolution should be as high as possible.

In those cases, interactive Earth observation means that the user in the ground station receives video images from the satellite and is able to steer the pointing direction of the camera platform via mouse control to the interesting event on the ground. As already mentioned, this is appreciated in applications where the target has not been identified clearly in advance, a search action is involved or a target has to be visually followed for a while.

The satellite is a cube of 32 cm × 32 cm × 32 cm and weighs 44.8 kg (Fig. 11).

The payload module contains two fore field cameras with low and medium resolution and a high resolution telescope with a focal length of 1 m, a pixel size of 8.3 μm and a GSD of 6 m. Each CCD-chip contains 752 × 582 pixels, and each camera can transmit video images in CCIR-standard and single digital images. The focal lengths of the fore field cameras are 16 mm and 50 mm, respectively (Table 3).

The micro-satellite DLR-TUBSAT worked well as a test satellite for interactive Earth observation. The special control features resulted in many informative and impressive passes. Fig. 12 shows the principle of the surveillance control.

This test satellite also delivered information on how to improve the performance. After some years of technological advances, the idea of DLR-TUBSAT was then also implemented in LAPAN-TUBSAT

Table 2
Calculated parameters of fire fronts of Fig. 9.

No	Eq. Fire temp. (K)	Eq. Fire area (ha)	Front length (km)	Energy release (MW)	Front strength (kW/m)
1	815	0.48	4	130	30
2	715	2.3	7.5	310	40
3	893	0.59	3	210	70
4	>670	<0.78	5	79	15
5	852	0.92	10	300	30
6	957	1.0	9	530	60
7	>690	<0.51	4	62	15
8	796	0.39	3	96	30

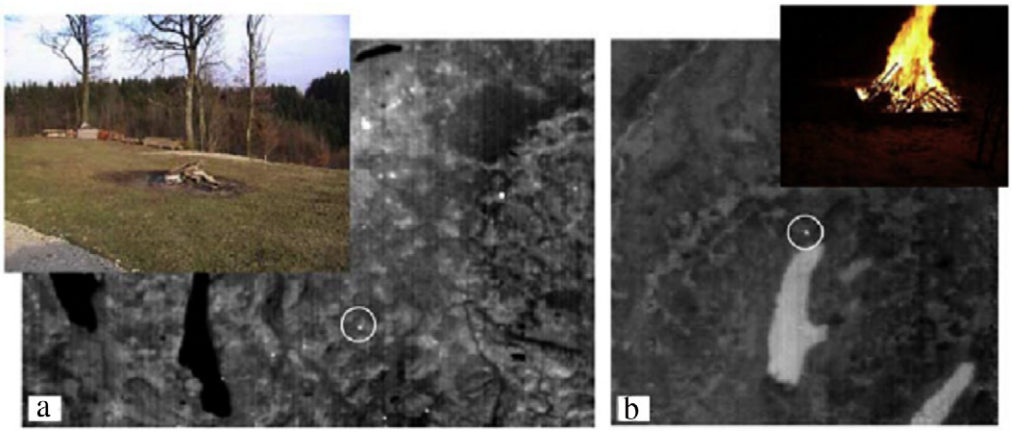

Fig. 10. Verification of the BIRD hotspot detection: (a) fire with an area of 12 m^2 detected in daytime; (b) controlled fire with an area of 4 m^2 detected at night (Zhukov et al., 2005).

Table 3
DLR-TUBSAT system data.

Mass	44.81 kg
Volume	32 cm^3 × 32 cm^3 × 32 cm^3
Communication	VHF/UHF-band, 1200 baud, 3.5–5 W
	S-band, 125 kbaud BPL coded or video transmission, 3.5 W
Attitude control	3 wheel/gyro-units ACS 202 (TUB), magnet torquer
Payload	f = 16 mm, GSD = 375 m
	f = 50 mm, GSD = 120 m
	f = 1000 mm, GSD = 6 m

which was jointly built by TUB and the Indonesian Space Agency LAPAN (Renner and Buhl, 2008). It was launched on 10 January 2007 by an Indian PSLV and is still jointly operated from ground stations in Rumpin (Indonesia) and Berlin.

5.2. Distributed space systems

In the last few decades, spacecraft mass and power have been reduced to limit costs and risks leading to shorter development times, the possibility to fly up-to-date technology, and frequent re-flight. Besides such advantages at the single spacecraft level, it must be considered that cost reduction also opens the possibility to fly multiple spacecraft systems. A system of platforms can replace a monolithic system very effectively with the advantage of ideally substituting the concept of failure with one of graceful degradation. In addition, a system of platforms can achieve a performance unachievable with a monolithic approach. Although the distributed system concept can be in principle applied to both large and small spacecraft, it is naturally coupled with small satellites for a number of reasons: system cost, easy replacement of a failed satellite (both cost and delivery time) and the possibility to gradually update flying technologies (which is generally an issue for large space systems).

Thinking about distributed space systems, we may distinguish between different systems based on the distance between the satellites and the requirements concerning the control of their distance (Gill, 2008). With this approach we get the following categories:

- Constellations.
- Formations.

The concept of formation flying of satellites is frequently confused with that of satellite constellations. In the following subsections we will distinguish between both concepts using the definitions of NASA GSFC:

Fig. 11. DLR-TUBSAT.

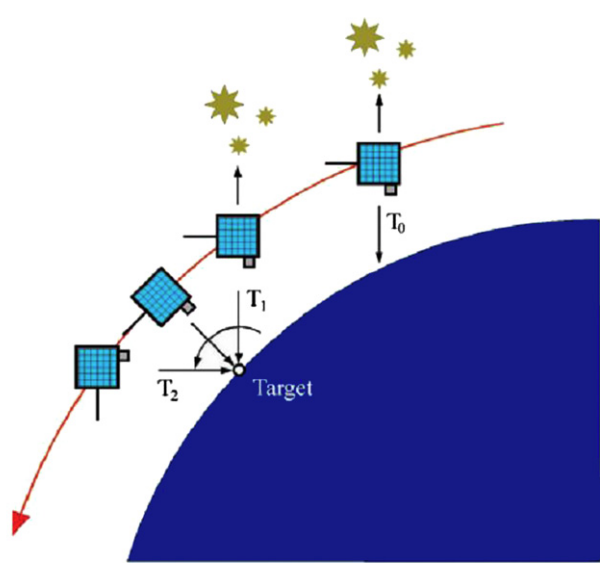

Fig. 12. Principle of surveillance control. The satellite turns from an inertial oriented flight (fixed attitude related to the stars, T0–T1) to an interactive target pointing mode (T1–T2).

– A constellation is composed of two or more spacecraft in similar orbits with no active control by either to maintain a relative position.
– Formation flying involves the use of an active control scheme to maintain a relative position.

From the application point of view, the aim of a constellation is generally related to coverage enhancement and, therefore, to the reduction of repetition time of observation of the same ground target and required time to achieve global coverage. In practice, a constellation consists of identical spacecraft whose orbits are designed to adequately cover the globe (or part of it). The constellation product is just the sum of the spacecraft products.

On the other hand, formations tackle the issue of the synergic use of payloads on board different platforms with no advantage in terms of time resolution. Thus, they enable additional products with respect to the ones offered by a single spacecraft. At the other end of the spectrum, one can imagine a formation implementing a distributed payload, with no product at all delivered by the payloads on board the single spacecraft. The clear advantage of such an approach is the possibility to fly very large sensor apertures in space.

In general, the manifold application areas of space-borne systems for Earth observation and surveillance cover very large ranges in terms of ground resolution, spectral resolution and time resolution (see Section 3). Some applications need to be implemented using distributed satellite systems like constellations or formations.

5.2.1. Constellations

A constellation can be defined as "a collection of satellites of similar design performing similar functions in similar orbits at the same time" (Wertz, 2001). There are a number of key issues which dominate constellation design tradeoffs. The altitude, the minimum elevation angle, the inclination and the constellation pattern determine the coverage, the main performance parameter, and the number of satellites, the main cost driver. Thus, the primary purpose of the mission planners is designing a constellation composed of the smallest number of satellites to achieve the mission requirements. The first constellation studies focused on the identification of the minimum number of satellites required to provide continuous Earth coverage: Easton and Brescia (1969) concluded that at least six satellites are required, while Walker (1970) concluded that five satellites are able to cover continuously the entire globe. Draim (1985, 1987), removing the circular orbit assumption, reduced the numbers to four and three, but at the cost of an extremely high altitude. The streets of coverage (SOC) method was developed by several authors (e.g. Lüders, 1961) with worse results with respect to Walker's method. Traditional satellite constellation design methods are associated with global or zonal, continuous (single or manifold) coverage under the typical assumption that payload field of view is a wide circular cone whose axis is along the satellite local vertical.

Constellation applications were first focused on navigation and communications, which justify these assumptions which, on the other hand, are extremely limiting when remote sensing applications are pursued. In fact, the first constellation implementations were related to navigation (NAVSTAR GPS, GLONASS and the future Galileo) with MEO orbits and communications (Iridium, Globastar) with LEO orbits.

Remote sensing sensor apertures are typically much smaller and their fields of view are shaped differently, e.g. Synthetic Aperture Radars (SAR) require side-looking geometry. Another feature of remote sensing payloads is that they need to pass over the imaged areas (e.g. for optical sensors with linear CCDs and for SAR), while navigation and communication payloads access all of the conical field of view at the same time. Thus, constellation design methods are not applicable to remote sensing missions and, in any case, lead to a huge number of required satellites when it is considered that MEO altitudes are typically not acceptable for such payloads. In practice, high resolution remote sensing missions are not feasible under the assumption of continuous global coverage. That is why remote sensing constellations have been proposed and implemented only later, without any utilization of the classic design methods.

Considering radar sensors, the COSMO/SkyMed constellation (Caltagirone et al., 2001; Battazza et al., 2008) consists of four 1900 kg satellites (three already in orbit) in the same orbital plane, symmetrically distributed with a 90° relative shift in latitude, on a frozen, sun synchronous, dawn–dusk, repetitive orbit which fly X-band SARs. Its capabilities will also increase when integration with the Argentinean SAOCOM constellation of two L-band SARs (CONAE, 2009) is operational in the SIASGE system. After COSMO/SkyMed (with a planned replacement after mission lifetime with COSMO 2), ESA will also fly Sentinel-1 (Davidson et al., 2006; Attema et al., 2007), a dual satellite system using C-band SAR flying in a dawn–dusk sun synchronous orbit. The Sentinel-1 spacecraft mass is 2300 kg, greatly reduced with respect to the multi-payload platform of Envisat (8140 kg). In addition, Canada is developing the Radarsat C constellation (Séguin and Eddy, 2006) with a maritime surveillance application as its driving requirement. In particular, Radarsat-C will have 3 satellites in the same orbital plane with 120° mean anomaly shift. Orbits are dawn–dusk sun synchronous at an altitude of 592 km. Radarsat-C satellites are intended to be 1300 kg mass, with a great improvement with respect to the 2200 kg of Radarsat-2. Finally, the SAR-Lupe constellation is an operational German military system relying on 5 satellites in LEO, with apogees at 505 km, perigees at 468 km and an inclination of 98.2°. The satellites are distributed over three orbital planes. It must be noted that at the century's turn the approach to spaceborne SAR has clearly changed from single satellite missions to constellation missions.

With reference to electro-optical payloads, two examples are given here to show their potential: DMC and RapidEye. These constellations are not specifically designed to support global change studies, but they may be used for such studies. At least they may be used as a model for future space-borne systems dealing with global change aspects.

The SWARM mission, other than using just one orbit plane (DMC and RapidEye) to provide wide surface coverage of imaging, is dedicated to the magnetosphere and uses 3 orbital planes: two of them slightly different to measure the east–west gradient flying side-by-side, while the third spacecraft flies at a higher altitude to support the separation of the different sources of the magnetic field.

5.2.1.1. Disaster monitoring constellation DMC.

As already mentioned, small satellites provide a unique opportunity for affordable constellations. In this respect, small satellites can do things that are not practical with large satellites. There are already two generations of DMC. The first generation of DMC may serve as an example for a constellation of five small satellites (Kramer, 2002). The standard spacecraft weighs 88 kg, of which 19 kg is payload. DMC has a GSD of 32 m and a swath width of 600 km (Landsat: GSD = 30 m, swath width = 185 km). It provides daily coverage of the Earth (Da Silva Curiel, 2005). The five satellites (AlSat-1, BILSAT-1, NigeriaSat-1, UK-DMC-1, Beijing-1) from five countries have been launched with three COSMOS launchers into the same orbit. Fig. 13 illustrates the coverage capability of four DMC satellites of the first generation (Da Silva Curiel et al., 2005). The DMC second generation is a planned follow-up to the DMC first generation with improved performance based on new technologies and it is already partially in operation.

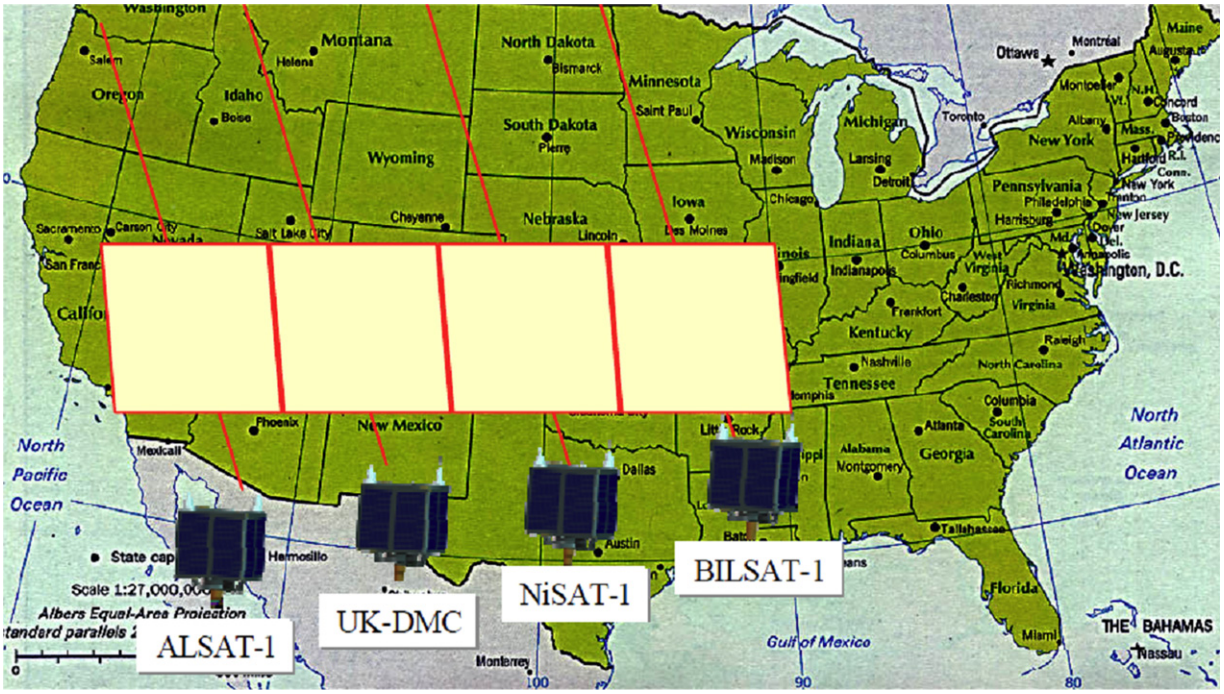

Fig. 13. Adjacent swath capability of DMC (Da Silva Curiel, 2005).

5.2.1.2. RapidEye.

RapidEye is a commercial multispectral Earth observation mission of RapidEye AG of Brandenburg, Germany, that includes a constellation of five mini-satellites (DLR, 2008) (Fig. 14). The mission provides high-resolution multispectral imagery along with an operational GIS (Geographic Information System) service on a commercial basis. The objectives are to provide a range of Earth observation products and services to a global user community. The RapidEye sensor images five optical bands in the 400–850 nm range (VIS and NIR) and provides 6.5 m pixel size at nadir. It provides products for the following applications:

- Agricultural producers (farmers): crop monitoring and mapping, yield prediction;
- Agricultural insurance: provision of regularly updated field maps to help insurers assess insurance contracts and claims by providing quick and reliable information about damaged areas;
- Cartography—satellite based maps (scale 1:25,000), ortho photos, DEM (Digital Elevation Model) generation.
- Disaster assessment, 3-D visualization.

The service spectrum of the RapidEye mission includes a guaranteed daily revisit, global coverage, product delivery to the customer within 24 h, the possibility of dedicated programming, the capability of direct transmission and imagery transfer within hours, a global digital database of "orthomaps" of 1:25,000 scale and DEMs of 20 m × 20 m resolution. The service permits also the merging of multi-temporal imagery with information from other sources.

The five RapidEye Earth observation satellites (1 m × 1 m × 1 m for a 150 kg mass) were launched on a single Russian Dnepr rocket from the Baikonur Cosmodrome in Kazakhstan on 29 August 2008. They have been deployed in orbits at an altitude of 630 km and placed equally spaced in a single sun-synchronous orbit to ensure consistent imaging conditions and a short revisit time. The satellites follow each other in their orbital plane at about 19 min intervals (Fig. 14). They are scheduled for a seven-year lifetime. The constellation approach in a single orbital plane permits a cumulative swath to be built up (the spacecraft view adjacent regions of the ground, with image capture times separated by only a few minutes). A revisit time of one day can be obtained anywhere in the world (±70° latitude) with body pointing techniques. The average coverage repeat period over mid-attitude regions (e.g., Europe and North America) is 5.5 days at nadir. The RapidEye system can access any area on Earth within one day and cover the entire agricultural areas of North America and Europe within five days. The swath width is 80 km and the maximum scene length per orbit is 1500 km.

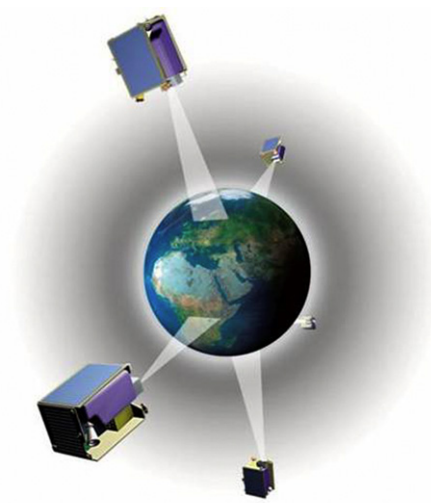

Fig. 14. RapidEye constellation of five satellites (DLR, 2008).

5.2.1.3. Swarm.

The ESA Swarm mission (Merayo et al., 2007) was selected as the 5th mission in ESA's Earth Explorer Programme in 2004 and is developed under the Living Planet Programme. The mission aims to provide the best ever survey (spatial and temporal accuracy) of the geomagnetic field and its temporal evolution that will lead to new insights into the Earth system by improving our understanding of the Earth's interior and its effect on geospace, the vast region around the Earth where electrodynamic processes

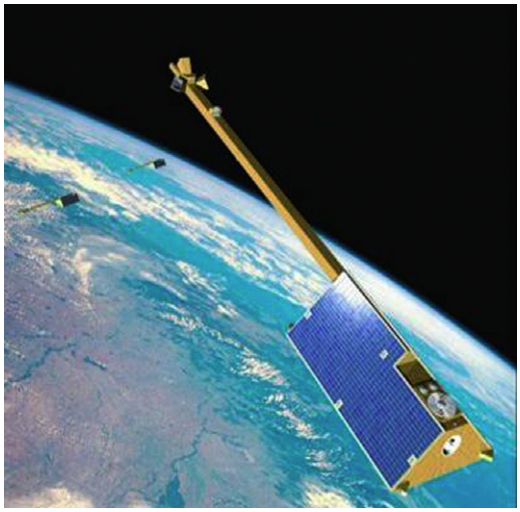

Fig. 15. Swarm satellite (image courtesy ESA).

are influenced by the Earth's magnetic field. The constellation of three spacecraft (Fig. 15), scheduled for launch in 2010, integrates two spacecraft flying side-by-side at lower altitude (450 km initial altitude), thereby measuring the East–West gradient of the magnetic field, and the third one flying at higher altitude (530 km). High-precision and high-resolution measurements of the strength, direction and variation of the magnetic field, complemented by precise navigation, accelerometer and electric field measurements, will provide the observations that are required to separate and model the various sources of the geomagnetic field. This results in a unique "view" inside the Earth from space to study the composition and processes of its interior. It also allows us to analyze the Sun's influence within the Earth system. In addition practical applications in many different areas, such as space weather, radiation hazards, navigation and resource management, will benefit from the Swarm concept (Merayo et al., 2007).

The scientific payload consists of the following instruments: Vector Field Magnetometer (VFM), which is co-mounted together with a stellar compass for determining the components of the magnetic field very accurately; Absolute Scalar Magnetometer (ASM), which is used primarily for calibrating absolutely the vector field magnetometer; Electrical Field Instrument (EFI) and Accelerometer (ACC). The scientific payload packages will fly on three 300–400 kg spacecraft, each of them eight meters long (see Fig. 15).

5.2.2. Satellite formations

The first applications of formations were proposed by astronomers to obtain optical interferometry able to cancel out starlight in order to identify distant planets, which imposes a very demanding (nm order) requirement on satellite relative position knowledge and control (e.g. Duren and Lay, 2002) whose fulfillment has been studied in deep space orbits.

Then, formation missions have been studied and proposed for Earth observations. Bistatic/Interferometric SAR missions have been studied in Italy (BISSAT/SABRINA: D'Errico et al., 1996; D'Errico and Fasano, 2008), France (Cartwheel: Massonnet, 2001) and Germany (Tandem-X: Krieger et al., 2008) following the first ideas developed during TOPSAT mission studies (Zebker et al., 1994). All these studies foresaw tandem missions, with the exception of Cartwheel which, although working with two satellites at a time, was based on three satellites (at least) flying in coordination with Envisat. Tandem-X is the only study which became a real mission and was launched on 21 June 2010 atop a Russian Dnepr rocket from the Baikonur Cosmodrome in Kazakhstan. More recently, ESA is studying formation-based SAR missions (Sephton et al., 2010): TOPOLEV is a dual satellite configuration with Ku-band radar; C-PARAS is a three-satellite configuration with receiving-only radar relying on the Sentinel-1 radar signal.

Multistatic Synthetic Aperture radar missions have been recently highlighted in the US (Goodman et al., 2002; Goodman and Stiles, 2003) to explore the possibility to integrate very large sparse apertures. Such an approach was also explored during the TechSAT21 mission studies (Burns et al., 2000; Steyskal et al., 2001; Steyskal and Schindler, 2003) at the Air Force Research Laboratory. One of the objectives of TechSat-21 was the assessment of space-based, sparse-array apertures used to synthesize a large radar antenna. In fact, the radars were designed to work in three modes: synthetic aperture radar imaging, moving target indication and geo-location. TechSat21 working modes were characterized by strong interactions among different radars. Multistatic SAR was also studied at CNES with the SAR train concept (Aguttes, 2005) which was introduced to solve the minimum SAR antenna constraint by splitting a wider antenna over a given number of smaller antennas. A similar approach (multi-aperture azimuth sampling) is followed in Li et al. (2005, 2006) and Gebert et al. (2006) for the proposed, virtually filled aperture. That is to say, that the SAR distributed system simulates a virtual monostatic SAR antenna. N smaller antennas work at an undersampling pulse repetition frequency, but signals acquired at different positions eventually by different antennas are coherently combined to gather the signal of the virtual antenna. Once the virtual SAR signal is reconstructed, standard SAR processing can be applied.

With reference to electro-optical payloads, fewer studies and mission proposals are documented in the literature. The concept of large filled apertures was for example exploited at JPL (Mettler et al., 2005). In this work a large aperture space telescope orbiting in GEO orbit (geostationary) is dealt with. The innovative concept is that the system, named GEOTEL, is made up by separate free flying optical modules, thus allowing for high resolution Earth observation. In particular, it is composed of six bodies. The primary mirror, also called the Primary Electrostatic Membrane Reflector (25 m diameter), is a gossamer concept, thus it is based on a future technology. Gossamer structures consist of thin film membranes, non-rigid in nature but assuming a prescribed figure thanks to an electron beam scanning its surface and controlling embedded actuators, and are considered one of the future breaking technologies for several potential applications (Bekey, 2003). Severe constraints exist on real time relative motion knowledge (nanometer range and sub-microradian bearing) and control (sub-mm). The architecture poses many challenges in the fields of centralized and decentralized sensing, estimation and control. Different advanced metrology systems (RF and optical) are used. Apart from the sunshade for the primary mirror, the other elements have a weight of 100 kg (primary mirror), 25 and 50 kg. A similar concept is proposed by Wertz (2004). In this case, the primary mirror is made up by 962 m diameter close free-floating optical modules which all move in a near GEO orbit. The primary mirror is protected from the Sun by an orbiting sunshade, which is 170 m wide and 70 m high, and which is made of Kapton. It keeps the primary mirror at a temperature of 40 K. In the area of sparse optical apertures, Kong et al. (1999) proposed an Earth imager from GEO to explore the use of an interferometer based on a distributed satellite system to image terrestrial objects.

Large sparse/filled apertures (Fig. 16) are breaking concepts that can be enabled by distributing the sensors over a large number of spacecraft. The cost (apart from the capability to design and operate a distributed payload) is the necessity of implementing autonomous formation flying, which implies the capability to accurately measure in real or almost real time their relative position, velocity and attitude and developing control laws and ad

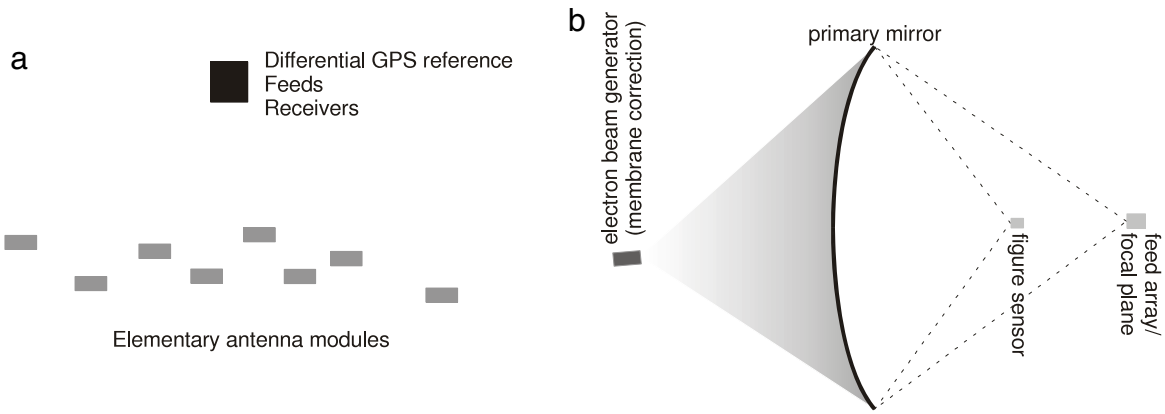

Fig. 16. General concepts of large sparse (a) and filled (b) apertures (adapted from Bekey, 2003). Sparse apertures require differential positioning measurements to correct the received signals before combination, whereas filled apertures require figure sensors to provide geometric error information to be fed back into the electron beam generator (to achieve membrane geometric correction). A free flying focal plane (or feed array) is then needed for large mirrors (or antennas).

hoc hardware to properly control relative status vectors. As a rule of thumb, the accuracy required in formation flying (measurement and, depending on the application, control) is related to sensor wavelength. Thus, moving from optical to microwave payloads means moving from nanometer to centimetric or larger accuracy is clearly a great technological advantage reducing by several orders of magnitude the required accuracy: e.g. rough positioning accuracy reduces from nanometers to centimeters or tenths of centimeters (in the case of P-band). This is likely the reason why radar missions have reached a more detailed design status and concentrated on LEO, whereas electro-optical mission studies moved to GEO to achieve a less perturbed gravity environment. Nevertheless, it is not guaranteed that a given optical line of sight can be achieved by an interferometer without using forced motion for the collectors and/or the combiner.

Although there seems to be a high degree of risk in developing formation-based missions, the formation concept is widely considered as a unique tool to perform missions otherwise impossible. In particular, formation flying is thought to be a revolutionizing technology with an impact on remote sensing missions (Leitner, 2004). For the studies to become missions, further developments are needed at the bus and payload level, involving both theoretical, hardware, and software issues. Microtechnology and nanotechnology will certainly play a fundamental role in satellite miniaturization, thus contributing to cost reduction.

Besides the need for technological development, some remote sensing missions are already flying utilizing the formation flying concept, such as GRACE (better described in the following) and A-TRAIN. A-Train, basically a constellation mission with some formation aspects, integrates different payloads on board different satellites (OCO, Aqua, Aura, PARASOL, Cloudsat, and CALIPSO) whose combined data allow us to study climate change issues. The formation flying aspect is particularly related to CALIPSO and Cloudsat whose time separation is controlled within 15 s of each other, so that both instruments will view the same cloud area at nearly the same moment. Such formation flying requirements came out from the scientific requirement to observe the same clouds (whose lifetime is often less than 15 min) at different wavelengths. A-Train is characterized by large, small (CALIPSO: 635 kg), and micro-satellites (PARASOL: 120 kg). Similarly, the NASA morning train consists of four satellites sharing the same track: Terra, SAC-C, EO-1, and Landsat-7. They are organized into two satellite pairs, whose satellites are separated by a 1 min time shift. As in A-Train, the morning train utilizes both large and small satellites, with EO-1 and SAC-C weighing around 500 kg.

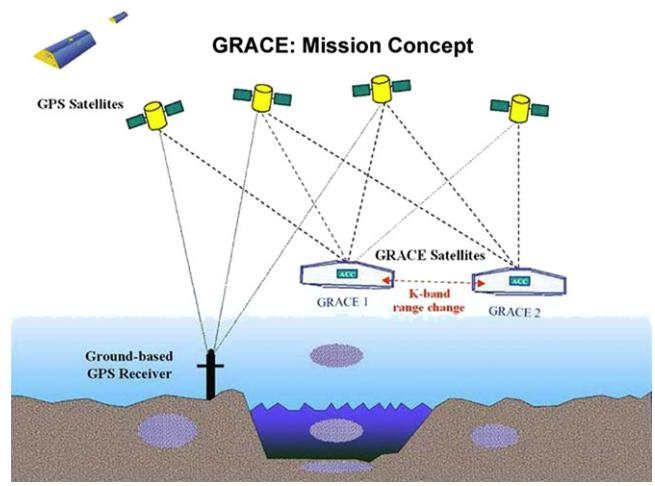

Fig. 17. GRACE—mission concept (GFZ, 2008).

5.2.2.1. GRACE.

The GRACE mission (Fig. 17) was selected as the second mission under the NASA Earth System Science Pathfinder (ESSP) Program in May 1997 as a joint partnership with Deutsches Zentrum für Luft- und Raumfahrt (DLR) in Germany. Launched in March 2002 for a five-year lifetime, it is still accurately mapping variations in the Earth's gravity field, thanks to the two identical spacecraft flying at about 220 km apart in a polar orbit 500 km above the Earth (CSR, 2008). The gravity field of the Earth is variable in both space and time, and is an integral constraint on the mean and time variable mass distribution in the Earth. The science data from the GRACE mission will be used to estimate global models for the mean and time variable Earth gravity field approximately every 30 days. The science data from the GRACE mission consists of inter-satellite range change measurements, and the accelerometer, GPS and attitude measurements from each satellite (Fig. 17).

The gravity variations that GRACE will study include: changes due to surface and deep currents in the ocean; runoff and ground water storage on land masses; exchanges between ice sheets or glaciers and the oceans; and variations of mass within the Earth. Another goal of the mission is to create a better profile of the Earth's atmosphere. The results from GRACE will make a considerable contribution to the goals of NASA's Earth Science Enterprise, Earth Observation System (EOS) and global climate change studies.

5.2.2.2. Distributed active radar remote sensing and small satellites.

Synthetic Aperture Radar satellites have been characterized by a large mass reduction in the last twenty years (considering

that the ERS-1 mass was about 2.4 tons), which is not a trivial result considering that SAR payload capabilities have been greatly improved with respect to, as an example, ERS-1 radar: electronic beam steering, different operating modes (e.g. stripmap, spot, ScanSAR), multi-polarization, antenna phase center split, etc. In part, such a successful mass reduction has been obtained by a change in satellite development philosophy: from the ENVISAT multi-payload platform (about 8 tons) to dedicated platforms. As examples, COSMO-SkyMed is about 2 tons, while the TerraSAR-X mass is about 1.2 tons. Greater results have been obtained in the cases were radar payloads have been simplified:

- SAR-Lupe relies on a platform of 4 m × 3 m × 2 m but with a mass of about 770 kg. One of the major cost (and mass) savings is the use of a single beam parabolic SAR reflector antenna instead of the more common active phased arrays (e.g. COSMO-SkyMed and TerraSAR-X). Thus, a highly efficient TWT (Travelling Wave Tube) and a low-loss high-gain antenna assure efficient utilization of the spacecraft antenna. The cost is the capability of stripmap mode only and of spot mode, but relying on the spacecraft attitude maneuver.
- TecSAR (Sharay and Naftaly, 2006) is realized by an even smaller platform (launch mass of 300 kg). Again, the SAR antenna is a parabolic reflector but, in this case, electronic beam steering is enabled by a multi-beam feed array. The TecSAR pointing capability is also enhanced by the capability of rolling the satellite.

Contrary to a widespread feeling in the past, active remote sensing has become compatible with the small satellite concept. Nonetheless, greater and wider advantages can be foreseen in the area of formation-based missions, when active radar systems can be complemented by receiving-only radars. Such results, achieved at a study level and not at an implementation level, are quite common:

- the first BISSAT study (D'Errico et al., 1996) which identified a 600 kg parasitic spacecraft coupled with ENVISAT-1;
- Cartwheel (Massonnet, 2001) was based on a 100 kg baseline micro-satellite concept;
- C-Paras (Sephton et al., 2010) relies on small satellites;

An additional strategy to reduce the power consumption problem consists of sharing the illuminating function among different spacecraft: TechSAT21 utilized 5 microsatellites in the 100 kg class which was also the baseline assumption in the SAR train study (Aguttes, 2005). A multi-satellite configuration to enable large SAR sparse apertures in principle should additionally enhance such advantages at the spacecraft level.

6. Conclusions

The general fact given in Section 2.1, especially the features depicted in Fig. 1, indicate clearly: the smaller the satellite the less the cost and response time of a space mission. This is a strong motivation to try implementing small satellite missions.

Small satellite missions can be achieved by using different approaches and methods. One approach is to focus on a single task and use available off-the-shelf technology to build a small satellite system (bus and payload) for the intended remote sensing purpose. Another possible approach is to take full advantage of the ongoing technology developments leading to further miniaturization of engineering components, and development of micro-technologies for sensors and instruments which allow designing dedicated, well-focused high-performance Earth observation missions.

Since the advent of modern technologies, small satellites using off-the-shelf technologies or missions focused on specific physical phenomena have also been perceived to offer an opportunity for countries with a modest research budget and little or no experience in space technology, to enter the field of space-borne Earth observation and its applications. Small satellite technology is a major means to bring within the reach of every country the opportunity to operate small satellite Earth observation missions and utilize the data effectively at low cost, as well as to develop and build application-driven missions. It provides the opportunity to conduct or participate in Earth observation missions using small, economical satellites, and associated launchers, ground stations, data distributions structures, and space system management approaches. The actual performance parameters of small satellites given in Section 4 show that small satellites are competitive with previous larger satellites with respect to spatial and spectral resolutions. Constellations of small satellites provide daily global coverage of the Earth's surface.

The examples in Section 5 are indicators of small satellite capabilities. For instance, the micro-satellite BIRD provides a better hot spot detection and assessment capability than the orbiting large satellites which are not optimized for this application.

DLR-TUBSAT exemplifies that the high agility of small satellites allows easy interactive control for surveillance purposes and observing local short term dynamics. Such a low-cost system can be launched with very low response time.

Constellations like DMC-1 and RapidEye provide already daily coverage of the globe with GSDs sufficient for many applications. With constellations of satellites injected into different orbital planes, the revisit time can be shortened or the time resolution can be increased.

Formations of small satellites like GRACE can register even minor changes in the mass distribution of the Earth and in the magnetic fluxes.

The main obstacles small satellites face are the restrictions in mass and volume. That leads to restrictions in power consumption and data rates because of the physical size of components like solar panels and antennas. This applies also to application-specific instrument components like high resolution optics and radar antennas. But also in those fields of technology and application the development results are encouraging.

There are plans on the table to complement active radar systems by receiving-only radars based on small satellites down to the micro-satellite class. Multi-satellite configurations to enable large SAR sparse apertures should additionally enhance the applicability of small satellites.

Another important obstacle caused by restricted mass and volume is the focus on selected physical phenomena from a small satellite platform. But, as pointed out before, this may even turn out to become an advantage if we think of building complex Earth observation systems using a complex suite of instruments accommodated on a distributed small satellite platform system. In the case of the malfunction of one system, it can be easily and cost-effectively replaced. NASA's A-Train and morning train include aspects of formation flying with small satellites making use of the distributed platform system approach. So, in the future the number of applications which will be covered by small satellite missions using active and passive remote sensing systems will be steadily increasing. This is a necessary process because the importance of remote sensing is increasing since we have to face an increasing world population and decreasing resources at the same time asking for an ever increasing number of cost-effective space systems to provide quality and timely information.

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